

QAOA parameter setting

25.11.2025

Summary and Agenda

Summary

Currently investigating hardware results to cement some of the conclusions from approximate training. The QAOA Training pipeline has been extended.

Agenda

- Daniel: general update.
- Stephan
- Conrad: high-level quality summary.
- Anurag & David.

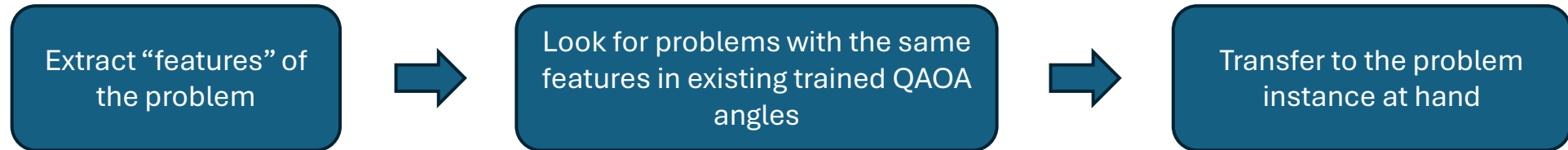
General update: training pipeline

- Extension to database centric training methods:
Define a trainer framework that can load data, process it, and produce QAOA parameters based on the data

https://github.com/qiskit-community/qaoa_training_pipeline/pull/32 Merged

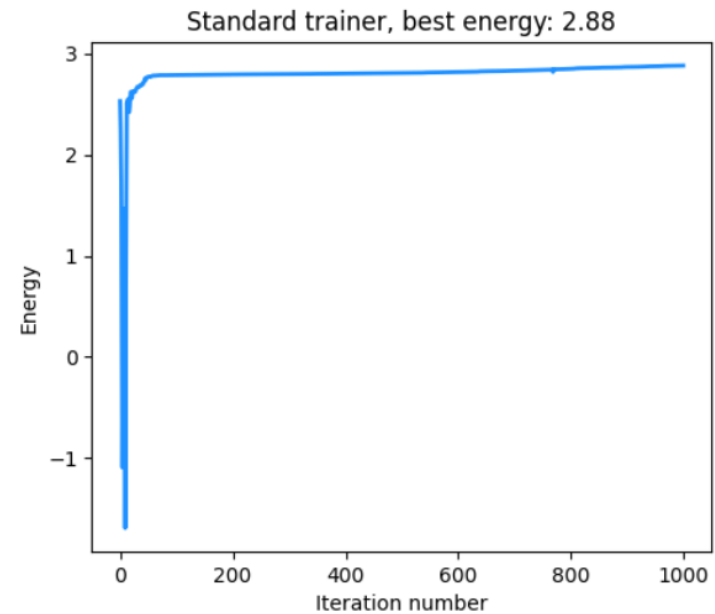
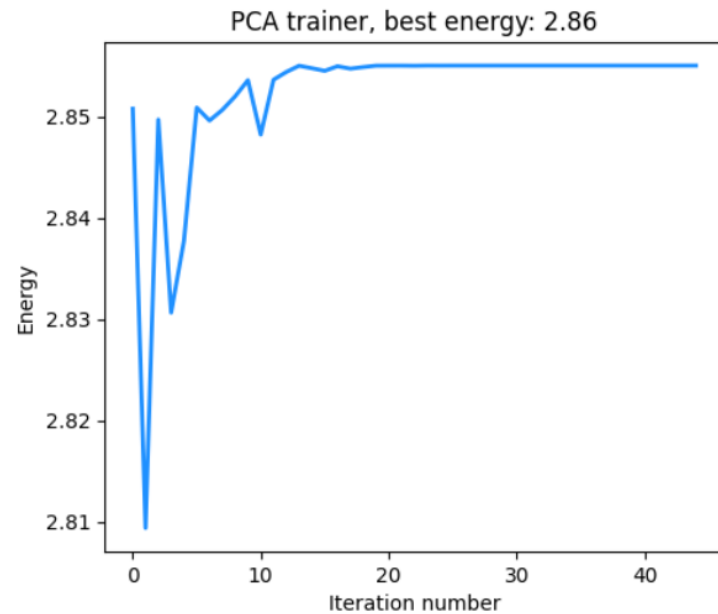
Supports:

- PCA-QAOA: principal components are identified from pre-trained QAOA parameters, transferred and then re-trained.
- Transfer of QAOA angles



General update: training pipeline

- Extension to database centric training methods:
PCA trainer (based on Parry & McMinn 2025)

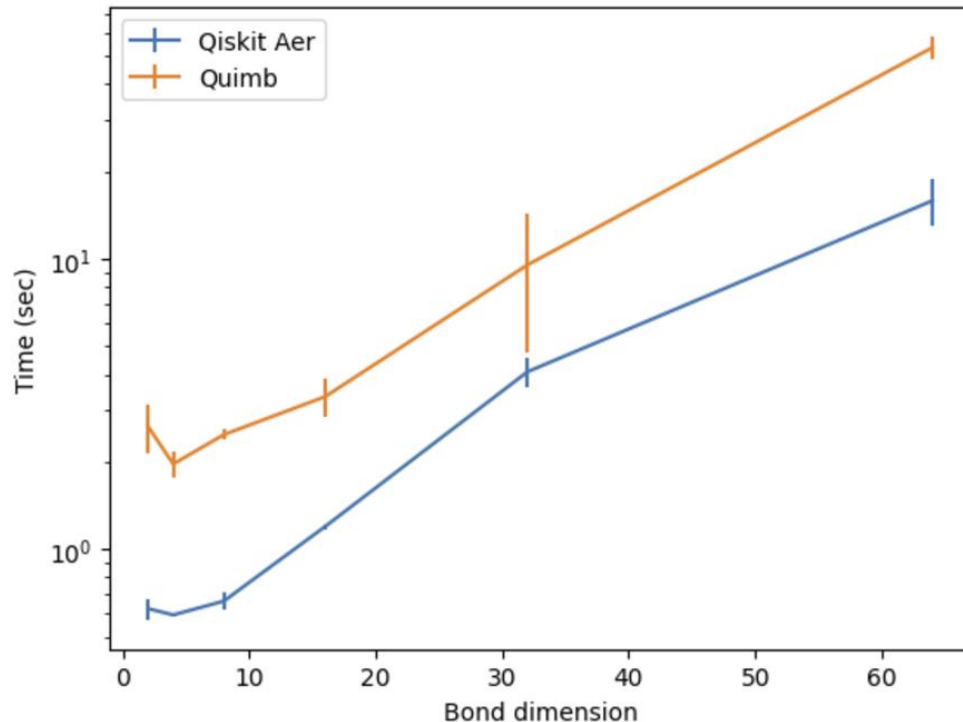


General update: training pipeline

- PR for Qiskit Aer MPS interface

https://github.com/qiskit-community/qaoa_training_pipeline/pull/36

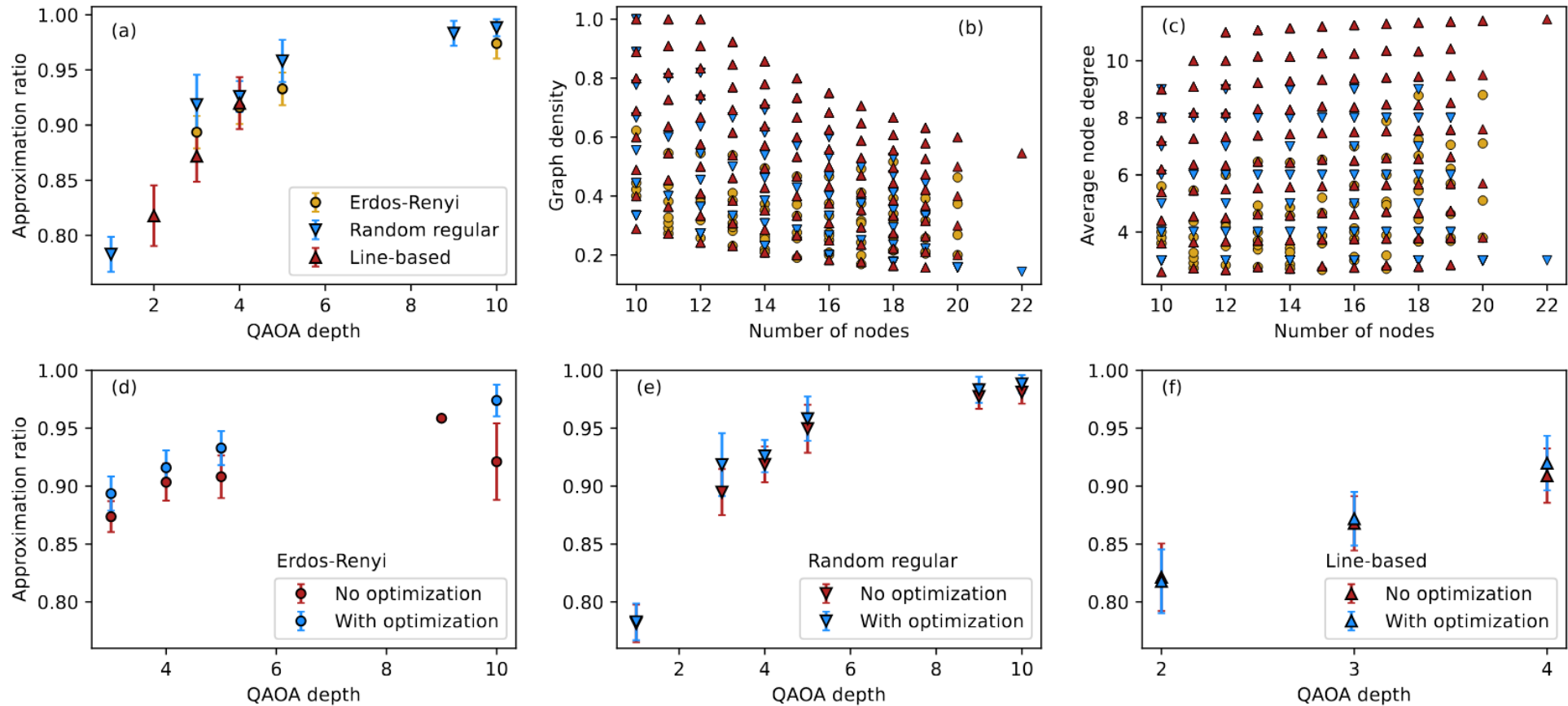
Qiskit Aer MPS appears to be faster than the Quimb implementation.



Four RR3 graphs per data point,
depth-one QAOA.

General update: distilling best practices

- INTERP and Fixed-Angles work the best so far.

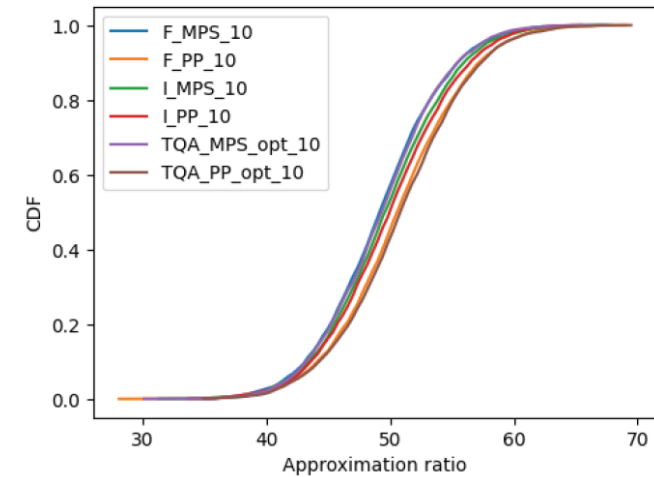
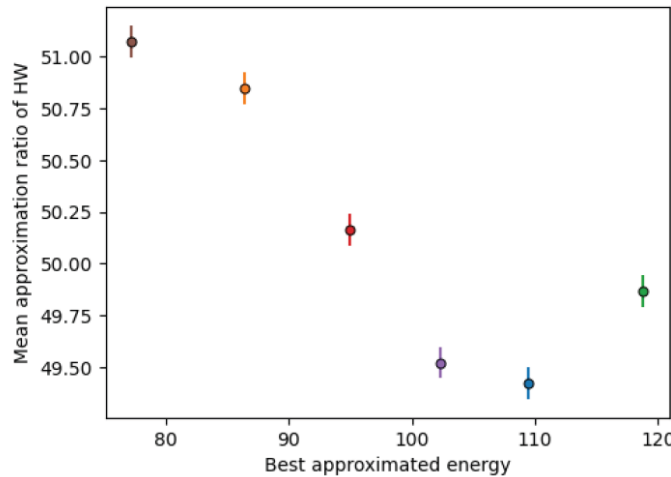
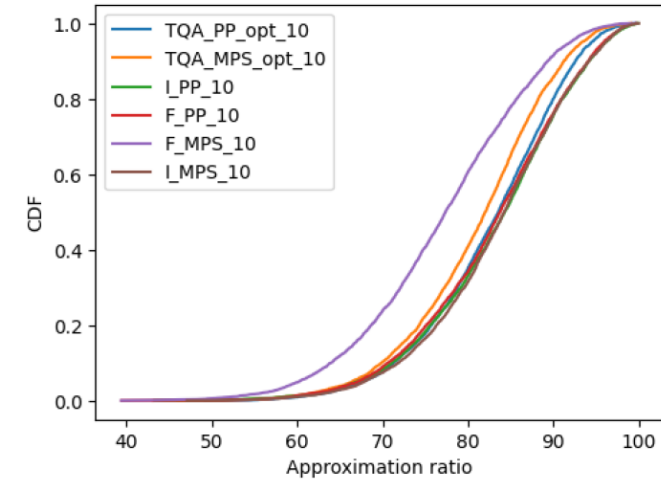
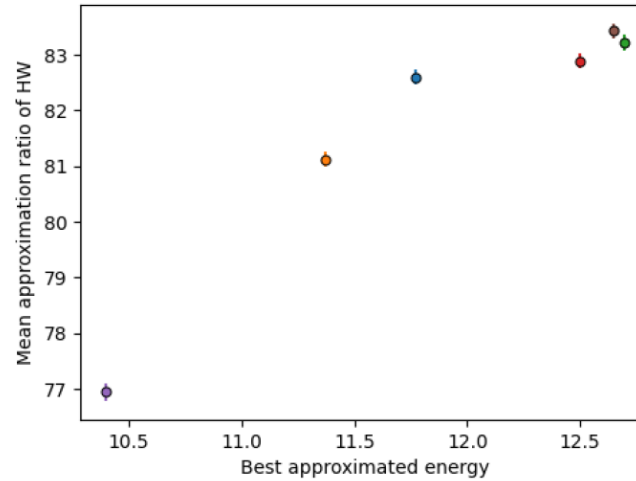


General update: hardware results

- Preliminary data

Top: 39-node heavy-hex graph with edge weights.

Bottom: 100 node graph with 10 swap layers.



Next steps

- More progress on the overleaf is needed, I'll start some skeleton text to point people to the right location.
- Validation: more hardware data is needed.
- More benchmarking
 - Utility-scale problems that can be run on hardware (ongoing)
 - Transfer of parameters from StateVector simulations to utility-scale (ongoing)
- Start to distill best practices from the observations.
- Scoring of the different methods
 - High-level quality summary
 - Resource cost analysis
 - Validation on quantum hardware
- Look at MIS and LABS
- Next call?

Notes

- Note

Next call

- In three weeks from now: Tuesday 25th November

Discussion of 25.11.2025

- Notes